Development of a Real-Time Hand Tracking Visualizer

1. Project Goal

Develop an interactive 3D application for visualizing hand-tracking data from different sources in real time.

The application should support both direct sensor measurements (e.g., Ultraleap hand-tracking sensors) and 3D hand poses predicted by machine-learning models.

The goal is to create a modular visualization tool that can be reused for debugging, comparing and evaluating different hand-tracking pipelines.

2. Functional Requirements

A. Hand Skeleton Visualization

- Display 3D joint positions and skeletal connections.
- Clearly distinguish left and right hands.
- Support multiple hands and independent tracking sources.
- Display hands in a consistent world coordinate system.
- Allow free rotation, zooming and navigation of the 3D scene.

B. Multiple Input Sources

- Direct live tracking from Ultraleap sensors.
- Predicted 3D joint positions from an external hand-pose-estimation model.
- A standardized input interface that can support additional sources.
- Optionally, playback of previously recorded hand-tracking sequences.

C. Hand Mesh Visualization

- Provide a toggle to enable or disable a 3D hand mesh.
- Fit or animate an anatomical hand model based on the incoming 3D joints.
- Support left- and right-hand models.
- Allow switching between skeleton-only, mesh-only and combined visualization.

D. Multi-Sensor Support

- Support independent streams from multiple sensors.
- Transform tracking data into a common coordinate system.
- Investigate calibration methods for determining relative sensor positions and orientations.

3. Initial Research Tasks

Before implementation, investigate the following topics:

Hand Representations

- Joint-based hand representations and skeleton topologies.
- Differences between Ultraleap, OpenXR and common 21-keypoint representations.
- Joint positions versus joint rotations and bone transformations.
- Anatomical and parametric hand models.

Hand Meshes and Fitting

- Rigged hand meshes and skeletal animation.
- MANO and other parametric hand models.
- Inverse kinematics and optimization-based mesh fitting.
- Real-time feasibility, computational costs and fitting quality.

Calibration and Coordinate Systems

- Calibration methods for multiple hand-tracking sensors.
- Temporal synchronization and handling of asynchronous measurements.

4. Expected Deliverables

Minimum viable implementation

1. A working interactive 3D viewer.
2. Live hand skeleton visualization from an Ultraleap sensor.
3. Support for externally supplied hand-pose predictions.
4. Correct left/right-hand identification and joint mapping.
5. Toggleable hand-mesh visualization.
6. Documented software architecture and setup instructions.

Optional Extensions

- Multiple sensors with spatial calibration.
- Overlay of predictions and sensor measurements.
- Tracking confidence, joint errors or latency visualization.
- Recording, playback and export functionality.
- Advanced mesh fitting or multi-sensor fusion.

5. First Assignment

For the next meeting:

1. Prepare a short overview of common hand representations and available hand models.
2. Compare at least two approaches for fitting a hand mesh to 3D joints.
3. Investigate how Ultraleap provides tracking data and how multiple devices can be supported.
4. Propose an initial software architecture and visualization framework.
5. Identify technical challenges and open questions.
6. Make timeline for yourself and show it to me